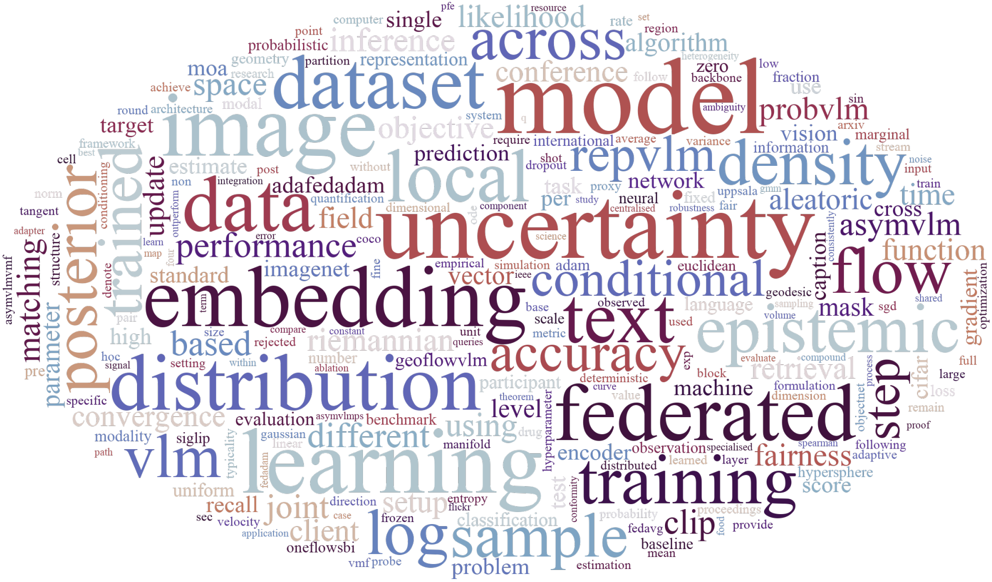
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fedavg
clip
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log
sample
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mean
vmf
probe
estimation
problem

Robust Learning
from Distributed and Heterogeneous Data

Li Ju

Division of Scientific Computing
Department of Information Technology
Uppsala University

June 4, 2026

When textbook assumptions break...

Typical data assumptions in machine learning

- Centralised: collected into one place
- Mono-modal: a single, well-formed channel
- Complete: all relevant information is observed

When textbook assumptions break...

Typical data assumptions in machine learning ... and what the real world says:

- Centralised: collected into one place
  - $\Rightarrow$ data are generated and kept at the edge, i.e., inherently distributed.
- Mono-modal: a single, well-formed channel
  - $\Rightarrow$ intelligence demands reasoning across heterogeneous modalities.
- Complete: all relevant information is observed
  - $\Rightarrow$ data are partially observed, i.e., any variable may be missing.

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Typical data assumptions in machine learning ... and what the real world says:

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ML methods should remain **robust** when these idealised assumptions break.

Towards robust ML

distributed
where data live

Federated Learning

Paper I, II

heterogeneous
how data are shaped

Uncertainty Quantification for Vision-language Models

Paper III, IV

incomplete
any subset observed,
any subset inferred

Simulation-Based Inference

Paper V

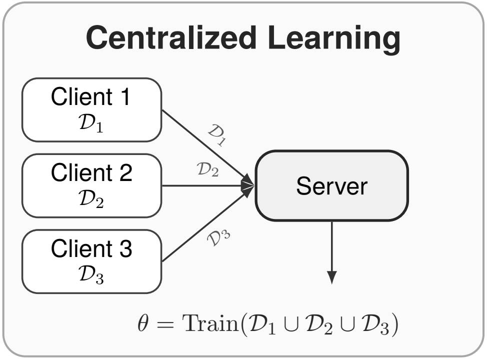
Paper I: An empirical study

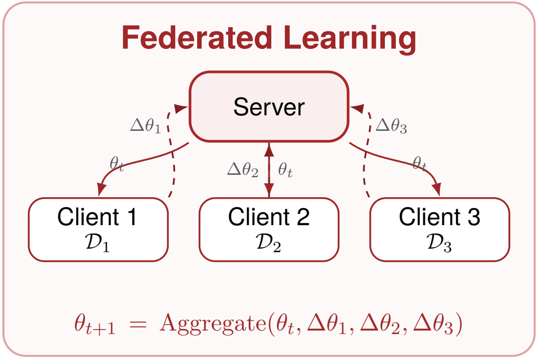
Federated learning for predicting compound mechanism of action based on image-data from cell painting [1]

[1] Ju et al., *Artificial Intelligence in the Life Sciences*, 2024

What is Federated Learning?

$$\theta^\star = \arg\min_\theta R(\theta, \mathcal{D})$$





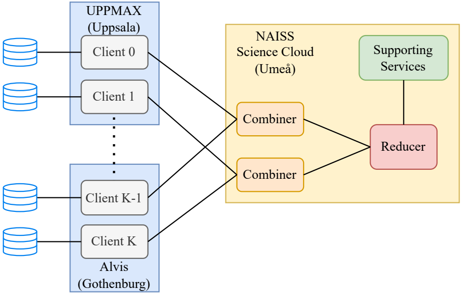
Moving the model to the data, instead of moving the data to the model.

FL for drug discovery: an empirical study

Task: Predict a compound's Mechanism of Action (MoA) from fluorescence *cell-painting* images with a neural network.

Why FL: Pharmaceutical data are locked in silos (due to intellectual property, regulation, etc).

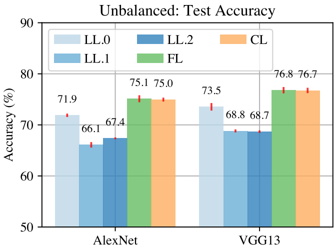
Three scenarios: Uniform (i.i.d.) → Unbalanced (size skew) → Non-IID (one client specialises in a subset of MoAs).



Implemented with FEDn, distributed across Uppsala, Gothenburg, and Umeå.

Three findings

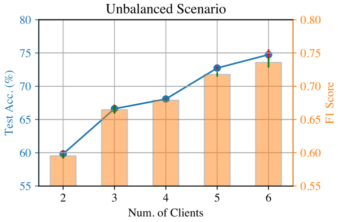
Finding 1



CL ≈ FL > Local models

Collaborate via FL.

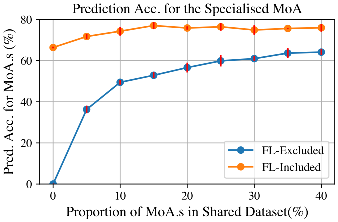
Finding 2



More clients ⇒ better model

Stay engaged.

Finding 3



Specialised client helps everyone

Specialised & general both benefit.

Federated learning works for MoA prediction — Non-IID data is not a big problem.

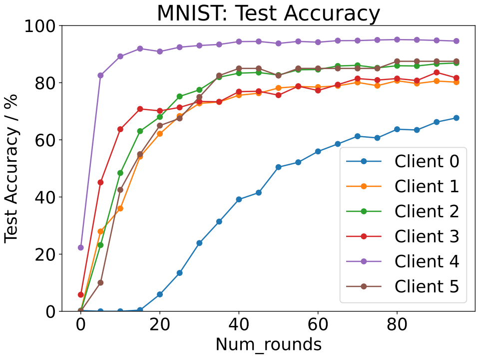
Paper II: Fairness for Federated Learning

Accelerating fair federated learning: Adaptive federated adam [2]

[2] Ju et al., *IEEE Transactions on Machine Learning in Communications and Networking*, 2024

Fairness problem

The global model can look good on average while being unusable for some clients.



Fairness problem: discrepancy of per-client model performance in FL.

Existing approach: Q-Fair FL

Standard FL minimises the sum of per-client risks; **Q-Fair FL**[3] raises each to power $q+1$:

$$\underbrace{\theta^\star = \arg\min_{\theta} \sum_{k=1}^{K} R_k(\theta)}_{\text{Standard FL}} \quad \Longrightarrow \quad \underbrace{\theta^\star = \arg\min_{\theta} \sum_{k=1}^{K} R_k^{q+1}(\theta)}_{\text{Q-Fair FL},\ q \geq 0}$$

- $+$: Proven fairness guarantee.
- $-$: Converge slow: 2–5$\times$ slower than `FedAvg`.
- $-$: Incompatible with accelerated federated optimizers, i.e., `FedOpt`.

We want federated learning to be both fair and fast.

[3] Li et al., *International Conference on Learning Representations*, 2020

Analysis & reformulation

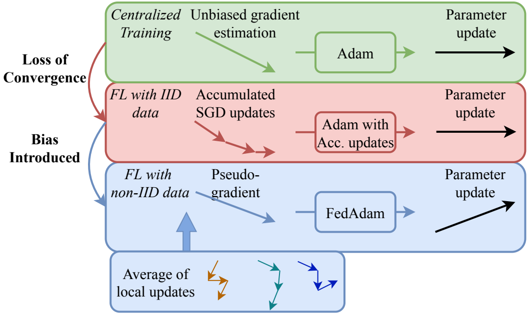
Key insight: The gradient of Q-Fair FL scales with the objective value, which can be arbitrarily small ⇒ slow convergence for first-order methods.

We propose a dynamic optimization formulation for fair FL, with two key properties:

- Same fairness guarantee as Q-fair FL (same stationary points).
- Gradient scales are preserved ⇒ compatible with `FedOpt` (e.g., `FedAdam`)[4].

[4] Reddi et al., *International Conference on Learning Representations*, 2021

From FedAdam to AdaFedAdam

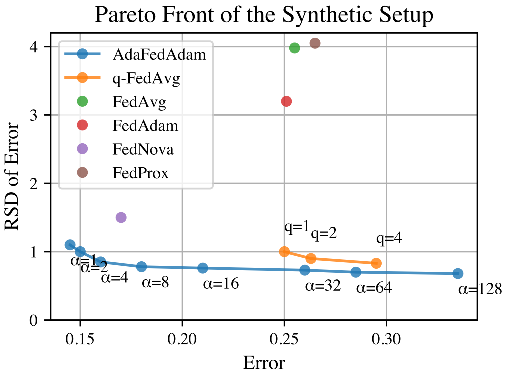


AdaFedAdam adds three fixes:

- Normalise local updates by ℓ_2 norm of local gradient.
- Estimate per-client confidence of the update direction.
- Use aggregated confidence to adapt server-side hyperparameters.

Empirical evaluation

- Faster: Up to 10$\times$ speedup over Q-Fair FL to target accuracy.
- More fair: Lower variance for prediction accuracy across clients.
- New Pareto frontier for fairness and accuracy trade-off in FL.



Paper III: Aleatoric uncertainty in vision–language models

Exploiting the asymmetric uncertainty structure of pre-trained vision–language models on the unit hypersphere[5]

[5] Ju et al., *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025

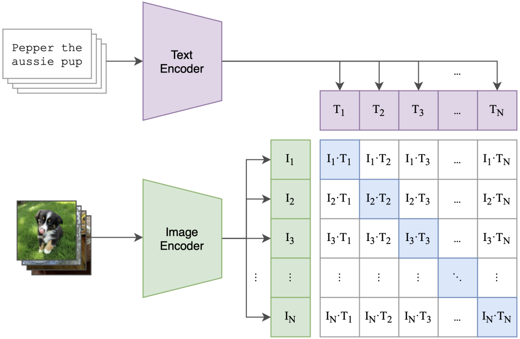
Vision-Language Models

Vision–Language Models (VLMs) embed images and texts into a shared representation space, so cross-modal retrieval reduces to a similarity lookup.

E.g., CLIP[6]

- Matched pairs: Pull together
- Mismatched pairs: Push apart

Cosine similarity is the distance metric
$\Rightarrow$ embeddings live $\mathbb{S}^{d-1}$.



[6] Radford et al., *International conference on machine learning*, 2021

Aleatoric vs. epistemic uncertainty

”I am uncertain about my answer...”

An output from a model can be uncertain for two distinct reasons:

- **Aleatoric** – inherent ambiguity in the data. Irreducible.
- **Epistemic** – the model’s ignorance. Reducible by more / better training data.

Paper III focuses on aleatoric uncertainty in pre-trained VLMs, and Paper IV on epistemic uncertainty.

Asymmetry in aleatoric uncertainty of VLMs

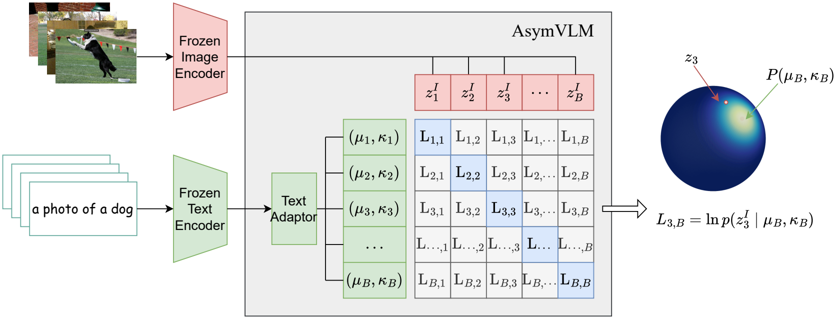
Hypothesis: both i2t and t2i are one-to-many mappings – not because images are ambiguous, but because texts are.

To capture this one-to-many mapping, we can equip the text embeddings with a probabilistic structure.

- **CLIP**: deterministic for both modalities – no notion of ambiguity.
- **ProbVLM**[7]: probabilistic for both modalities – overlooking the asymmetry .
- **AsymVLM (ours)**: asymmetrically probabilistic – probabilistic text, deterministic image.

[7] Upadhyay et al., *Proceedings of the IEEE/CVF international conference on computer vision*, 2023

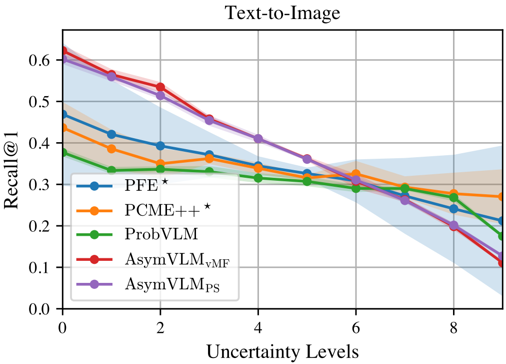
Building AsymVLM

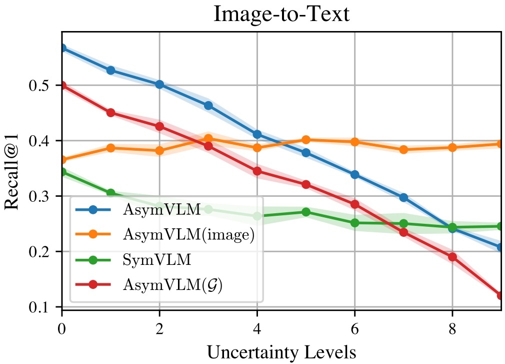


Geometry Respect.
Embeddings live on $\mathbb{S}^{d-1}$. CLIP text embeddings mapped to a directional distribution.

Maximum likelihood Principle.
Fit the text distribution so the matched image embedding has maximum likelihood.

Calibration and structural ablation





Effectiveness. Predicted uncertainty tracks retrieval error – the model is well-calibrated.

Ablation. Both asymmetric structure and directional distributions are both essential.

Paper IV: Epistemic uncertainty in vision–language models

Epistemic uncertainty quantification for pre-trained VLMs via Riemannian flow matching[8]

[8] Ju et al., *International conference on machine learning*, 2026

When the *model* is ignorant

AsymVLM captures aleatoric uncertainty (ambiguity) in the text, but there is no signal for model ignorance, i.e., epistemic uncertainty.

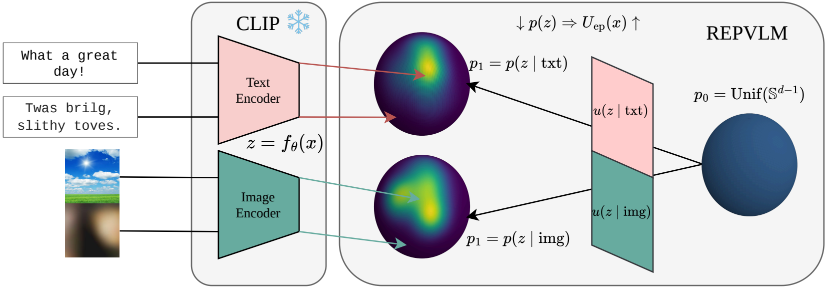
Existing options are unsatisfying:

- Deep ensembles – prohibitive for foundation-scale backbones.
- MC-Dropout – requires multiple forward passes.

We need a scalable, intrinsic measure of epistemic uncertainty for VLMs.

Embedding density proxies ignorance

Hypothesis. Epistemic uncertainty of x under VLM $f_\theta(\cdot)$ can be proxied by $-\log p(f_\theta(x))$.



The embedding $z \in \mathbb{S}^{d-1}$ of a familiar input lands in a dense region of the hypersphere; an unfamiliar input lands in a sparse region.

Embedding density estimation

Goal. Estimate $p(z)$ for $z \in \mathbb{S}^{d-1}$ that is

- high-dimensional (hundreds of dimensions),
- multi-modal (concepts cluster across the sphere),
- respects the spherical geometry.

Options

- Parametric directional distributions – too simple, can't capture multimodality.
- KDE / GMM – collapse in high dimensions, no geometry.
- RFM – scales to high dim, captures multimodality, and operates natively on the sphere.

Building RepVLM

Training – learn the vector field.

1. Design a probability path p_t from $p_0 = \mathrm{Uniform}(\mathbb{S}^{d-1})$ to $p_1 = p(z \mid c)$, generated by a velocity field $u_t(z, c)$ via the ODE

$$\dot{z}_t = u_t(z_t, c), \quad z_t \in \mathbb{S}^{d-1}, \;\; u_t(z_t, c) \in T_{z_t}\mathbb{S}^{d-1}, \;\; c \in \{\text{text}, \text{image}\}$$

   whose trajectories follow geodesics of the sphere.

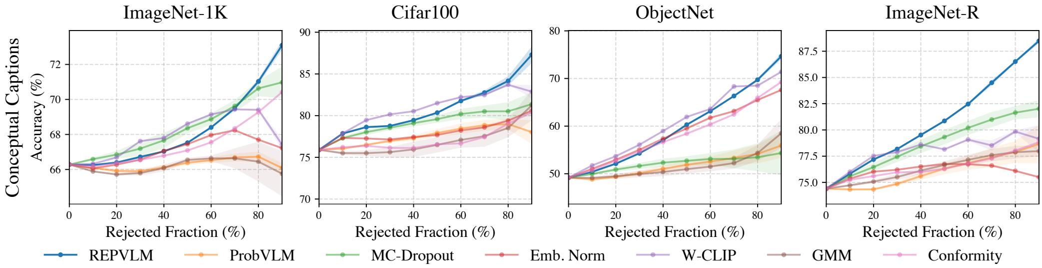
2. Approximate u_t with a neural network $v_t(z, c; \phi)$:

$$\min_{\phi} \; \mathbb{E}_{t,z} \left\| v_t(z, c; \phi) - u_t(z, c) \right\|^2.$$

Inference – evaluate density.

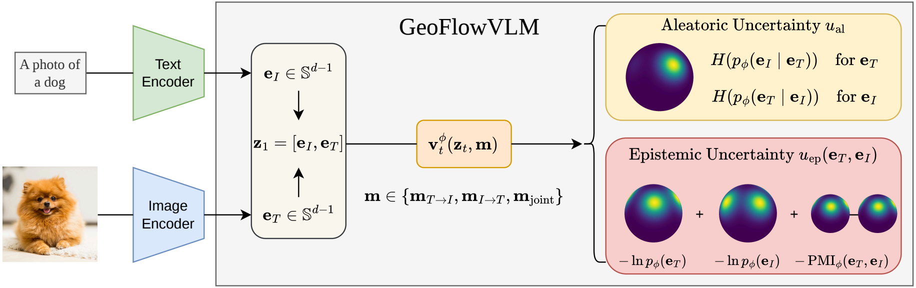
Integrate $\nabla \cdot v_t(\,\cdot\,; \phi)$ along the geodesic flow from $t = 0$ to $t = 1$ to recover $\log p_1(z \mid c)$ exactly.

Empirical evaluation



- Monotonicity between rejection rate and prediction error.
- Effective for OOD detection and automated data curation.
- Only $\sim 10\%$ the compute of MC-Dropout at inference.

Spoiler: GeoFlowVLM – a unified method



GeoFlowVLM[9]: A single masked Riemannian flow model over the joint embedding space that captures both aleatoric and epistemic uncertainty.

[9] Nautiyal et al., *arXiv preprint arXiv:2605.13352*, 2026

Paper V: All-in-one simulation-based inference

OneFlowSBI: One model, many queries for simulation-based inference[10]

[10] Nautiyal et al., *arXiv preprint arXiv:2601.22951*, 2026

What is simulation-based inference?

Scientific simulators $\theta \to y$ are widely used (epidemiology, climate, physics). But, when the likelihood $p(y \mid \theta)$ is intractable, the posterior

$$p(\theta \mid y) \;=\; \frac{p(y \mid \theta)\, p(\theta)}{p(y)}$$

cannot be evaluated directly.

Simulation-based inference (SBI)
Approximate $p(\theta \mid y)$ with simulated (θ, y) pairs via a surrogate model, e.g., neural networks.

Limitations of the standard SBI

It works, but...

Posterior-only is too narrow.

Beyond $p(\theta \mid y)$, we need

- likelihoods $p(y \mid \theta)$;
- marginals $p(y)$, priors $p(\theta)$;
- arbitrary conditionals – "what if θ_1 is fixed?"

Observations are partial.

- sensor dropouts;
- differing experimental setups across labs;
- corrupted or masked dimensions.

OneFlowSBI is one model that handles *any query*.

OneFlowSBI: one model, any query

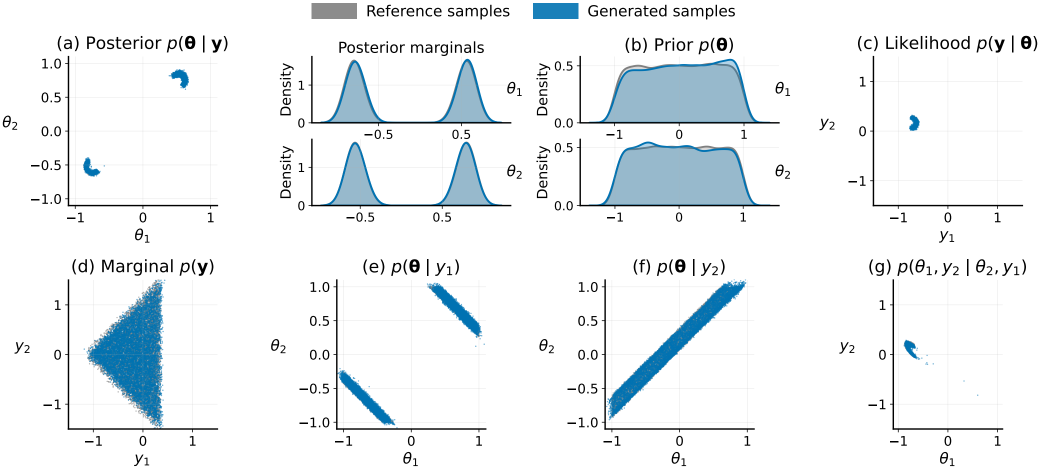
Training. Train one flow matching model over the joint state $\mathbf{z} = [\boldsymbol{\theta};\, \mathbf{y}] \in \mathbb{R}^{d_\theta + d_y}$. A binary mask $\mathbf{m} \in \{0,1\}^{d_\theta + d_y}$ tells the model what to condition on ($m_i = 1$, fixed) and what to generate ($m_i = 0$, transported).

Inference. Fix observed dims, draw $\varepsilon \sim p_0$ on the dims to be generated, integrate the masked ODE

$$\frac{d\mathbf{z}_t}{dt} = (\mathbf{1} - \mathbf{m}) \odot v_t^{\phi}(\mathbf{z}_t, \mathbf{m}), \quad \mathbf{z}_0 = \mathbf{m} \odot \mathbf{z}_{\text{obs}} + (\mathbf{1} - \mathbf{m}) \odot \varepsilon, \quad t \in [0, 1].$$

*Same machinery as GeoFlowVLM.

Querying OneFlowSBI



Pulling the three threads together

Federated Learning	robustness to client heterogeneity & unfairness
Vision–Language Models	robustness via uncertainty quantification
SBI	robustness to partial observations & arbitrary queries

Recurring methods

- Distributed optimisation – Papers I, II.
- Geometry-native probabilistic methods – Papers III, IV.
- Generative modelling via flow matching – Papers IV, V.

Future work

Goal: resilient, fair, and self-aware learning in the uncurated world.

- **At scale – Decentralised foundation models**: Push distributed optimisation to federated training of large multi-modal models.
- **Under generation – UQ for generative VLMs**: Quantify intrinsic hallucination for GPT-/Qwen-style generators.
- **In the loop – AI-driven scientific discovery**: Put OneFlowSBI in active-learning loops, proposing the next informative experiment.

Papers included in the thesis

I. **Ju L**, Hellander A, Spjuth O. Federated learning for predicting compound mechanism of action based on image-data from cell painting. *Artificial Intelligence in the Life Sciences* 5: 100098, 2024.

II. **Ju L**, Zhang T, Toor S, Hellander A. Accelerating fair federated learning: Adaptive federated adam. *IEEE Transactions on Machine Learning in Communications and Networking* 2: 1017–1032, 2024.

III. **Ju L**, Andersson M, Fredriksson S, Glöckner E, Hellander A, Vats E, Singh P. Exploiting the asymmetric uncertainty structure of pre-trained VLMs on the unit hypersphere. *Advances in Neural Information Processing Systems*, 2025.

IV. **Ju L**, Nautiyal M, Hellander A, Vats E, Singh P. Epistemic uncertainty quantification for pre-trained VLMs via Riemannian flow matching. *International Conference on Machine Learning*, 2026.

V. Nautiyal M, **Ju L**, Ernfors M, Hagland K, Holma V, Werkö Söderholm M, Hellander A, Singh P. OneFlowSBI: One model, many queries for simulation-based inference. *arXiv:2601.22951*, 2026.

Other papers during the PhD

- **Ju L**, Singh P, Toor S. Proactive autoscaling for edge computing systems with Kubernetes. *Proceedings of the 14th IEEE/ACM International Conference on Utility and Cloud Computing Companion (pp. 1-8)*, 2021.
- Nautiyal, M[*], **Ju, L**[*], Hellander, A, Vats, E, Singh, P. GeoFlowVLM: Geometry-Aware Joint Uncertainty for Frozen Vision-Language Embedding. *arXiv preprint arXiv:2605.13352*, 2026. ([*]Equal contribution)
- Zhang T, **Ju L**, Singh P, Toor S. InfoHier: Hierarchical information extraction via encoding and embedding. *arXiv:2501.08717*, 2025.
- Li S, Ngai EC, Ye F, **Ju L**, Zhang T, Voigt T. Blades: A unified benchmark suite for Byzantine attacks and defenses in FL. *IEEE/ACM Ninth International Conference on Internet-of-Things Design and Implementation (IoTDI) (pp. 158-169)*, 2024.
- Nautiyal M, Arranz Gheorghe S, Stefa K, **Ju L**, Sintorn IM, Singh P. PARIC: Probabilistic attention regularization for language-guided image classification from pre-trained VLMs. *arXiv:2503.11360*, 2025.